

Detailed 8-Week Curriculum

Week 1: Accelerated Foundry and Ontology Baseline

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Engineers who can:

- Build AI-ready enterprise data products
- Engineer unstructured-data and embedding pipelines
- Design agentic and multi-agent systems
- Implement governed operational writeback
- Create production Evals
- Manage LLM releases, observability and cost
- Extend Foundry using OSDK and backend APIs
- Architect secure, production-grade AIP solutions

Module 1: Accelerated Foundry Architecture

- Foundry, AIP and Apollo architecture
- Mapping cloud-native data architecture to Foundry
- Foundry projects, datasets and repositories
- Rapid overview of data connections
- Batch, incremental and event-driven data patterns
- Pipeline Builder advanced operations
- Complex incremental pipelines
- Branching and release-management patterns
- Late-arriving data handling
- Schema evolution
- Data contracts
- Pipeline dependency management
- Lineage and change-impact analysis

- Compute optimisation
- Development-to-production promotion

Module 2: Advanced Ontology Engineering

- Moving from relational models to Ontology models
- Objects, properties, links and interfaces
- Complex object relationships
- Stable object identity
- Derived properties
- Functions and business logic
- Action types
- State and event modelling
- Temporal relationships
- Ontology evolution
- Designing Ontologies for AI consumption
- Designing AI-readable metadata
- Ontology as a contract between data, applications and agents

Module 3: Enterprise Security Baseline

- Role-based access control
- Attribute-based access control
- Data markings
- Object-level security
- Function and action permissions
- Purpose-based access patterns
- Security propagation across pipelines and Ontology resources
- AI and agent permission boundaries
- Auditability of data and action access

Practical Lab: Insurance Ontology Baseline

Participants will:

1. Review preconfigured policy, customer, claim and payment data.
2. Build an incremental claims-transformation pipeline.
3. Configure schema and data-quality checks.
4. Implement branching and controlled promotion.
5. Create the following Ontology objects:
 - Policyholder
 - Insurance Policy
 - Claim
 - Claim Document
 - Payment
 - Adjuster
 - Investigation
6. Define complex object relationships.
7. Apply RBAC and ABAC controls.
8. Test restricted access to high-value and sensitive claims.
9. Review lineage and downstream dependencies.

Outcome: A secured, agent-ready insurance Ontology and production-oriented pipeline baseline.

Week 2: AIP Foundations and Enterprise LLM Architecture

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 4: Enterprise LLM Architecture

- LLM architecture relevant to enterprise solutions
- Tokens, context windows and model limitations
- Model-selection criteria
- Prompt architecture
- System, user and contextual instructions
- Zero-shot and few-shot techniques

- Structured output
- Prompt templates
- Context engineering
- Grounding LLMs with Ontology data
- Retrieval-augmented generation
- Model-routing patterns
- Handling hallucination and ambiguity
- Enterprise security considerations

Module 5: LLM Pipelines and AI FDE Workflows

- Designing reusable LLM pipelines
- Data preparation for LLM consumption
- Prompt and context pipelines
- Ontology-aware retrieval
- Function and tool integration
- AI Forward Deployed Engineering workflow
- Translating operational requirements into AIP use cases
- Rapid prototyping
- Testing with business users
- Moving from prototype to operational solution
- Building initial functional copilots
- Copilot limitations and escalation paths

Practical Lab: Insurance Claims Copilot

Participants will build a copilot that:

1. Accepts a claim identifier.
2. Retrieves the policyholder and policy.
3. Retrieves claim history.
4. Validates policy coverage.
5. Summarises the claim.

6. Highlights missing information.
7. identifies potential claim risks.
8. recommends the next investigation step.
9. provides grounded references.
10. returns structured output.
11. refuses unauthorised queries.
12. escalates low-confidence responses.

Outcome: A functional, Ontology-grounded claims copilot with structured and permission-aware responses.

Week 3: AIP Logic and Action-Oriented Writeback

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 6: Advanced AIP Logic

- AIP Logic architecture
- Typed workflow inputs and outputs
- Model and prompt blocks
- Ontology queries
- Retrieval blocks
- Function invocation
- Conditional routing
- Multi-step reasoning
- Deterministic and generative stages
- Tool-calling
- Error handling
- Retry and fallback logic
- Reusable AIP Logic functions
- Workflow testing and tracing

Module 7: Ontology Actions and Functions

- Moving beyond read-only chat
- Action-oriented AI
- Ontology Functions
- Action types
- Action parameters
- Precondition validation
- Postcondition validation
- Multi-object updates
- External-system writeback
- Idempotency
- Optimistic concurrency
- Duplicate-action prevention
- Compensation and rollback
- Audit logging

Module 8: Automate and Approval Workflows

- Trigger-based automation
- Scheduled automation
- Event-driven execution
- Multi-stage approvals
- Risk-based approval routing
- Human-in-the-loop workflows
- Human-on-the-loop workflows
- Policy-controlled autonomous execution
- Escalation and exception handling
- Separation of duties

Practical Lab: AI-Assisted Claim Approval and Writeback

The workflow will:

1. Retrieve the claim and policy.

2. validate policy coverage.
3. calculate an eligible claim range.
4. detect missing evidence.
5. assess fraud and operational risk.
6. recommend approval, rejection or investigation.
7. route the recommendation using a risk-based approval matrix.
8. update the claim through an Ontology Action.
9. initiate an external payment request for approved claims.
10. prevent duplicate payments.
11. create a compensation action if payment fails.
12. record the decision and evidence.

Outcome: An AI workflow that performs governed system updates rather than only generating recommendations.

Week 4: Unstructured Data and Insurance Document Intelligence

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 9: Unstructured Data Engineering

- Unstructured data in Foundry
- PDF, image, email, audio and document processing
- Media sets
- Physical and virtual media sets
- Document ingestion patterns
- OCR and text extraction
- Layout and table extraction
- Metadata extraction
- Document classification
- Sensitive-data identification
- Document lineage

- Incremental document processing
- Version and duplicate management

Module 10: Chunking and Embedding Pipelines

- Text cleaning and normalisation
- Fixed-size chunking
- Structural chunking
- Semantic chunking
- Chunk overlap
- Metadata-aware chunking
- Embedding-model selection
- Batch embedding generation
- Re-embedding strategies
- Semantic search
- Hybrid retrieval
- Access-aware retrieval
- Retrieval filtering through Ontology properties
- Measuring retrieval quality
- Managing embedding drift

Module 11: Insurance Document Intelligence

- Policy-document extraction
- Claim-form extraction
- Medical and damage-report processing
- Loss-assessor report analysis
- Invoice and receipt extraction
- Clause identification
- Coverage and exclusion retrieval
- Detecting missing documents
- Cross-document consistency checks

- Linking extracted facts to Ontology objects

Practical Lab: Automated Claim Document Intelligence

Participants will:

1. Load claim forms, policies, invoices and assessor reports.
2. Extract text, tables and metadata.
3. classify each document.
4. detect policy and claim identifiers.
5. remove duplicate content.
6. chunk documents using appropriate strategies.
7. generate embeddings.
8. create semantic and hybrid retrieval.
9. connect documents to Claim and Policy objects.
10. extract coverage limits and exclusions.
11. compare statements across documents.
12. identify missing or contradictory information.
13. test access restrictions.
14. measure retrieval accuracy.

Outcome: An insurance document-intelligence and embedding pipeline ready for AIP consumption.

Week 5: Agentic AI and Complex Multi-Agent Workflows

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 12: Agentic AI Architecture

- Copilots versus autonomous agents
- Agent goals, instructions and tools
- Planning and task decomposition
- Tool selection
- Ontology-aware agents

- Agent state and memory
- Event-driven agents
- Agent execution boundaries
- Stopping conditions
- Failure recovery
- Agent observability

Module 13: Multi-Agent Orchestration

- Supervisor and specialist agents
- Planner–executor pattern
- Router pattern
- Evaluator–optimizer pattern
- Reviewer–remediation pattern
- Agent-as-tool composition
- Sequential and parallel execution
- Typed agent handoffs
- Shared versus isolated context
- Conflict resolution
- Agent permission separation
- Controlling recursive behaviour
- Cost and latency considerations
- Multi-agent audit trails

Practical Lab: Multi-Agent Claims Investigation

Participants will implement:

Supervisor Agent

- Receives the claim.
- decomposes the investigation.
- delegates tasks.
- consolidates the final decision.

Policy Agent

- Verifies policy status.
- checks limits, exclusions and applicable clauses.

Document Agent

- analyses claim documents.
- identifies missing or contradictory evidence.

Fraud Risk Agent

- examines previous claims and behavioural indicators.
- produces a risk assessment.

Financial Agent

- validates invoices and calculates the eligible amount.

Evaluation Agent

- checks evidence, groundedness and policy compliance.
- accepts or rejects the proposed decision.

Action Agent

- writes the approved decision back to the Ontology.
- initiates authorised downstream actions.

Outcome: A multi-agent insurance workflow connected to the Ontology and capable of governed operational action.

Week 6: Production AIP, Evals and LLMOps Governance

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 14: AIP Evaluation Engineering

- Evaluation strategy
- Unit, component and end-to-end Evals
- Golden datasets
- Human-labelled evaluation data
- Deterministic scoring

- Model-based scoring
- Structured-output validation
- Retrieval relevance
- Context precision and recall
- Groundedness
- Hallucination measurement
- Tool-selection accuracy
- Function-argument correctness
- Action and writeback accuracy
- Task-completion rate
- Safety and policy adherence

Module 15: Production LLMOps

- Prompt and model versioning
- Evaluation versioning
- Model comparison
- Champion–challenger testing
- Regression testing
- Continuous evaluation
- Model and embedding drift
- Release gates
- Environment promotion
- Production feedback loops
- Rollback
- Incident management
- Audit and compliance evidence

Module 16: Observability, Capacity and Cost Governance

- Agent execution tracing
- Model-call telemetry

- Tool-call monitoring
- Failure and retry metrics
- Token utilisation
- Latency and throughput
- Model capacity controls
- Rate limits
- Usage quotas
- Cost attribution
- Budget thresholds
- Cost-aware routing
- Caching and context optimisation
- Runaway-agent prevention
- Production audit trails

Practical Lab: Production Eval and Release Gate

Participants will:

1. Create an insurance claims Eval dataset.
2. define expected retrieval results.
3. define expected tool calls.
4. validate structured outputs.
5. measure groundedness and hallucination.
6. test prompt-injection resistance.
7. evaluate writeback correctness.
8. compare two model or prompt versions.
9. define latency and cost limits.
10. create an automated release gate.
11. reject a failing version.
12. promote a passing version.
13. simulate production regression.

14. roll back to a stable version.

15. produce an Eval and governance scorecard.

Outcome: A controlled production-release process for models, prompts, agents, tools and writeback actions.

Week 7: AIP Extensibility, OSDK and Custom Integration

Duration: 4 hours

Theory: 2 hours | **Practical:** 2 hours

Module 17: Developer Console

- Developer Console architecture
- Custom application registration
- Backend-service configuration
- Selecting Ontology resources
- Generating scoped clients
- Application permissions
- OAuth client configuration
- Environment-specific configuration
- Managing application versions
- SDK documentation generation

Module 18: OSDK Engineering

- Ontology SDK architecture
- Python OSDK
- TypeScript OSDK
- Java and OpenAPI options
- Object queries
- Filtering and aggregation
- Relationship traversal
- Function invocation
- Ontology Action execution

- AIP Logic invocation
- Agent-session integration
- Permission-aware backend services
- SDK versioning and compatibility

Module 19: Backend, API and Authentication Engineering

- Backend-service architecture
- REST APIs
- Webhooks
- External functions
- External database integration
- Cloud-service integration
- Service-to-service authentication
- OAuth patterns
- Token lifecycle
- Secret and certificate management
- Network and egress policies
- Rate limiting
- Circuit breakers
- Idempotent operations
- API contract testing

Module 20: Streaming UX and Third-Party Extensions

- Streaming LLM response architecture
- Session and conversation lifecycle
- Partial-response handling
- Cancellation and timeout handling
- Backend-to-client event streaming
- Streaming error recovery
- Third-party extension registration

- Extension permission boundaries
- Approved dependency management
- Organisational policy enforcement
- Extension telemetry
- Security review and release approval

Practical Lab: OSDK-Based Streaming AIP Service

Participants will build a Python or TypeScript service that:

1. Authenticates through an approved client.
2. retrieves insurance Ontology objects.
3. traverses claim relationships.
4. invokes an AIP Logic function.
5. starts an agent session.
6. streams the agent's response.
7. calls an external insurance service.
8. validates the API response.
9. executes an Ontology Action.
10. prevents duplicate writeback.
11. handles timeout and authentication failure.
12. produces structured logs and metrics.
13. runs unit and contract tests.
14. packages the service for CI/CD.
15. validates the extension against organisational policy.

Outcome: A secure, streaming OSDK service connecting AIP, the Ontology and third-party systems.

Week 8: AIP Capstone and Strategic Presentation

Duration: 4 hours

Theory/Architecture review: 2 hours

Implementation and presentation: 2 hours

Module 21: Gap Analysis and Architecture Review

- Current-state workflow assessment
- Identification of high-value AIP use cases
- Data and Ontology readiness
- Unstructured-data readiness
- Model and retrieval requirements
- Agent and tool requirements
- Writeback and integration requirements
- Security and compliance gaps
- Operating-model implications
- Capacity and cost assessment
- Target-state architecture
- Delivery roadmap
- Risk and control matrix

Capstone Options

Teams may implement one of the following:

Option 1: Multi-Agent Claims Processing

- Document extraction
- Policy retrieval
- Fraud-risk analysis
- Financial validation
- Decision evaluation
- Governed claim writeback

Option 2: Insurance Underwriting Copilot

- Applicant and risk-context retrieval
- Document intelligence
- Rule and model orchestration
- Recommendation generation

- Human approval
- Underwriting-system writeback

Option 3: Autonomous Legacy Stored-Procedure Modernisation

- Retrieve an approved legacy stored procedure.
- discover its dependencies.
- reconstruct the business logic.
- map source and target schemas.
- generate the target implementation.
- generate tests and test data.
- execute source and target procedures.
- compare outputs.
- automatically correct discrepancies.
- run security and performance Evals.
- generate deployment and rollback scripts.
- deploy through a governed pipeline.
- monitor and automatically roll back failures.

Required Capstone Components

Every solution must include:

- Foundry data product
- Enterprise Ontology
- Structured or unstructured retrieval
- AIP Logic
- Agent or multi-agent workflow
- Ontology Action
- Governed writeback
- Eval framework
- Observability
- Security controls

- Cost and capacity controls
- OSDK or API integration
- Deployment and rollback strategy

Strategic Presentation

Each team will present:

1. Business problem and current-state gap
2. Proposed AIP architecture
3. Ontology design
4. Retrieval and grounding approach
5. Agent architecture
6. Writeback and approval model
7. Security architecture
8. Eval results
9. Capacity and cost strategy
10. Production deployment roadmap

Outcome: A reviewed enterprise AIP solution accompanied by implementation evidence and a strategic production roadmap.